

5004 Homework2 (Writing Part)

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Q1

Rewrite the linear system for iteration:

$$\begin{cases} x_1 = \frac{1}{8}(20 + 3x_2 - 2x_3) \\ x_2 = \frac{1}{11}(33 - 4x_1 + x_3) \\ x_3 = \frac{1}{12}(36 - 6x_1 - 3x_2) \end{cases}, \quad \mathbf{x}^{(0)} = (0, 0, 0)^T$$

(1) Jacobi: only uses value of previous step

Iteration 1 ($k = 1$):

$$\begin{cases} x_1^{(1)} = \frac{1}{8}(20 + 3(0) - 2(0)) = 2.5 \\ x_2^{(1)} = \frac{1}{11}(33 - 4(0) + (0)) = 3 \\ x_3^{(1)} = \frac{1}{12}(36 - 6(0) - 3(0)) = 3 \end{cases}, \quad \mathbf{x}^{(1)} = (2.5, 3, 3)^T$$

Iteration 2 ($k = 2$):

$$\begin{cases} x_1^{(2)} = \frac{1}{8}(20 + 3 \times 3 - 2 \times 3) = 2.875 \\ x_2^{(2)} = \frac{1}{11}(33 - 4 \times 2.5 + 3) \approx 2.364 \\ x_3^{(2)} = \frac{1}{12}(36 - 6 \times 2.5 - 3 \times 3) = 1 \end{cases}, \quad \mathbf{x}^{(2)} = (2.875, 2.364, 1)^T$$

(2) Gauss-Seidel: uses newest value

Iteration 1 ($k = 1$):

$$\begin{aligned}x_1^{(1)} &= \frac{1}{8}(20 + 3(0) - 2(0)) = 2.5 \\x_2^{(1)} &= \frac{1}{11}(33 - 4(2.5) + 0) \approx 2.0909 \\x_3^{(1)} &= \frac{1}{12}(36 - 6(2.5) - 3(2.0909)) \approx 1.2273\end{aligned}$$

Iteration 2 ($k = 2$):

$$\begin{aligned}x_1^{(2)} &\approx \frac{1}{8}(20 + 3(2.0909) - 2(1.2273)) \approx 2.9773 \\x_2^{(2)} &\approx \frac{1}{11}(33 - 4(2.9773) + 1.2273) \approx 2.0289 \\x_3^{(2)} &\approx \frac{1}{12}(36 - 6(2.9773) - 3(2.0289)) \approx 1.0041 \\ \therefore \mathbf{x}^{(2)} &= (2.9773, 2.0289, 1.0041)^T\end{aligned}$$

These manual calculation results were verified in the programming section.

Q5

Problem Statement

Consider the heat equation:

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$$

Show that the numerical scheme

$$\frac{U_j^{n+1} - U_j^{n-1}}{2\Delta t} = 2 \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{(\Delta x)^2}$$

is **second order** both in time and in space.

Proof via Local Truncation Error (LTE) Analysis

Let $u(x, t)$ denote the exact solution. The Local Truncation Error (LTE) is obtained by substituting the exact solution into the numerical scheme:

$$\tau_j^n = \frac{u(x, t + \Delta t) - u(x, t - \Delta t)}{2\Delta t} - 2 \frac{u(x + \Delta x, t) - 2u(x, t) + u(x - \Delta x, t)}{(\Delta x)^2}$$

We expand each part using Taylor series about the point (x, t)

Step 1: Taylor Expansion in Time

$$u(x, t + \Delta t) = u + \Delta t u_t + \frac{(\Delta t)^2}{2} u_{tt} + \frac{(\Delta t)^3}{6} u_{ttt} + \frac{(\Delta t)^4}{24} u_{tttt} + O((\Delta t)^5)$$

$$u(x, t - \Delta t) = u - \Delta t u_t + \frac{(\Delta t)^2}{2} u_{tt} - \frac{(\Delta t)^3}{6} u_{ttt} + \frac{(\Delta t)^4}{24} u_{tttt} + O((\Delta t)^5)$$

Subtracting:

$$u(x, t + \Delta t) - u(x, t - \Delta t) = 2\Delta t u_t + \frac{(\Delta t)^3}{3} u_{ttt} + O((\Delta t)^5)$$

Dividing by $2\Delta t$:

$$\frac{u(x, t + \Delta t) - u(x, t - \Delta t)}{2\Delta t} = u_t + \frac{(\Delta t)^2}{6} u_{ttt} + O((\Delta t)^4)$$

This is the centered difference in time, and its error is $O((\Delta t)^2)$.

Step 2: Taylor Expansion in Space

$$u(x + \Delta x, t) = u + \Delta x u_x + \frac{(\Delta x)^2}{2} u_{xx} + \frac{(\Delta x)^3}{6} u_{xxx} + \frac{(\Delta x)^4}{24} u_{xxxx} + O((\Delta x)^5)$$

$$u(x - \Delta x, t) = u - \Delta x u_x + \frac{(\Delta x)^2}{2} u_{xx} - \frac{(\Delta x)^3}{6} u_{xxx} + \frac{(\Delta x)^4}{24} u_{xxxx} + O((\Delta x)^5)$$

Adding and rearranging:

$$u(x + \Delta x, t) - 2u(x, t) + u(x - \Delta x, t) = (\Delta x)^2 u_{xx} + \frac{(\Delta x)^4}{12} u_{xxxx} + O((\Delta x)^6)$$

Dividing by $(\Delta x)^2$:

$$\frac{u(x + \Delta x, t) - 2u(x, t) + u(x - \Delta x, t)}{(\Delta x)^2} = u_{xx} + \frac{(\Delta x)^2}{12} u_{xxxx} + O((\Delta x)^4)$$

This is the standard second-order central difference in space, with error $O((\Delta x)^2)$.

Step 3: Combine and Compute LTE

Substituting both expansions into τ_j^n :

$$\begin{aligned} \tau_j^n &= \left[u_t + \frac{(\Delta t)^2}{6} u_{ttt} + O((\Delta t)^4) \right] - 2 \left[u_{xx} + \frac{(\Delta x)^2}{12} u_{xxxx} + O((\Delta x)^4) \right] \\ &= \underbrace{(u_t - 2u_{xx})}_{=0} + \frac{(\Delta t)^2}{6} u_{ttt} - \frac{(\Delta x)^2}{6} u_{xxxx} + O((\Delta t)^4) + O((\Delta x)^4) \end{aligned}$$

Since u is the **exact solution**, it satisfies $u_t = 2u_{xx}$, so the leading term vanishes. Therefore:

$$\tau_j^n = \frac{(\Delta t)^2}{6} u_{ttt} - \frac{(\Delta x)^2}{6} u_{xxxx} + O((\Delta t)^4 + (\Delta x)^4) = O((\Delta t)^2 + (\Delta x)^2)$$

Conclusion

The Local Truncation Error is $O((\Delta t)^2 + (\Delta x)^2)$, which proves that the scheme is **2nd order in time** (the $(\Delta t)^2$ term) and **2nd order in space** (the $(\Delta x)^2$ term).